

INDU: Spread of butterflies in Sverige

Lars Arvestad

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Introduction

The distribution of species changes over time, and in this project, you will write a program to study data from Artportalen, operated by the Swedish University of Agricultural Sciences (SLU), presenting data from SLU's Artdatabanken. Artdatabanken collects observations from both researchers and nature enthusiasts, covering all conceivable species. You will receive observations for some butterfly species, gathered over 20 years, and your program should generate diagrams illustrating various questions about the life and distribution of butterflies.

Take, for example, the Purple Emperor butterfly, which is found in large parts of Europe but until recently was sporadically seen only in the southernmost parts of Sweden. In recent years, it has started spreading northward. When did it start moving north? How far north does it travel? When in the year is it observed?

Your program should read observation data for any species, exported from Artportalen and saved as CSV files, and create diagrams for the species.

For general project requirements and grading criteria, refer to the general instructions at *INDU—Individual Project in Programming*.

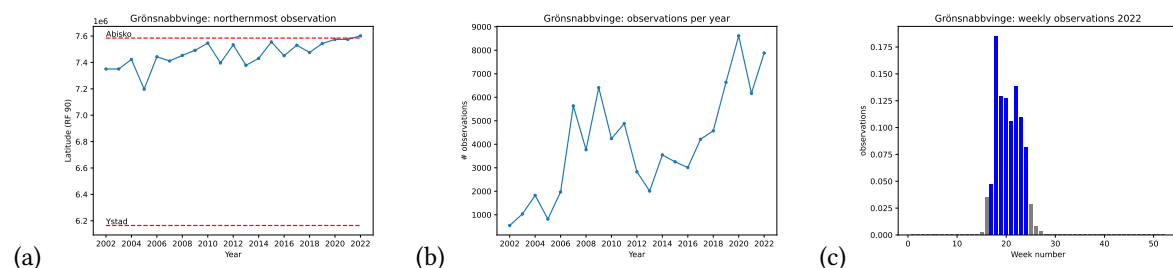


Figure 1: Examples of generated figures for the Green Swiftwing butterfly. (a) The diagram illustrates how the northernmost observations have changed over time. (b) The number of observations made per year. (c) The percentage of observations per week, with the weeks where 90 % of the observations are made highlighted in blue.

Assignment

You are to write a Python program that reads observation data from Artportalen (see section “Data” below) and analyzes how the distribution has changed over time. The program should primarily create figures using matplotlib which enables answering the following questions for each species under study:

- **How has the distribution of the species changed over the years?** Display in a figure how the northernmost observation has changed over the years. In other words, create a figure with years on the x-axis and the northernmost latitude for that year on the y-axis. Draw lines for Ystad and Abisko in the figure to clearly illustrate the latitudes.
- **How have the number of observations changed over time?** Show in a figure how many observations have been made per year in Sweden.
- **What time of the year is the species active?** Use the observations to assess during which time butterflies can be found in nature. The program should create a bar chart showing the percentage of observations per week (of the year’s observations), with week numbers on the x-axis, for the year 2022.

The program should also calculate and present in which period (given as “from week x to week y ”) 90 % of observations are made. This calculation is done by determining the week number at which the first 5 % and 95 % of observations have occurred. You can choose to either highlight with color in the diagram the weeks when the species is active (see Figure 1) or present it in text (printed in the console or the diagram).

With the help of your program and the data files you can download to your computer, it should be possible to address the following biological questions:

- Which species appear to have moved northward?
- Is the number of observations roughly constant over the years?
- Are there any patterns in the times when butterflies are active? Are there differences between species?

Since this is a programming course, you don’t need to answer these three questions, but consider them as a guide to the programming goals.

Criteria

- The program should be generic and capable of handling new data files from Artportalen (processed in the same way as the ones you receive).
- You are **not** allowed to use the pandas module.
- If problems arise during loading, an error message should be displayed, but the calculations should continue.
- You must not discard data that is difficult to interpret (e.g., with semicolons in the field Lokalnamn); use the csv module to read data.
- Axes should be labeled (e.g., ‘Year’ and ‘amount’) in your figures.
- Your program should answer all questions in a clear and concise manner.
- Your program should save the figures in PDF format with descriptive filenames.
- The printed figures should resemble those shown in Figure 1.

Rapporten

As described in the general instructions, the report should focus on the structure of the program, and the report should be a maximum of three A4 pages. However, in this assignment, you can include an appendix containing example figures with possible explanations and answers to the questions in the assignment.

Data

You will receive data in the ZIP file butterfly_data.zip under the project folder on the course website. The data files are essentially taken directly¹ from Artportalen, but some columns have been removed for privacy

¹Artportalen limits the number of exportable observations to 10,000, so some files have been combined from multiple queries.

reasons (e.g., observer's name) and to simplify the data files. However, the files still contain more data than you need, and several columns should be ignored. The format is "CSV UTF-8," exported from Excel, meaning that columns are separated by semicolons. These data *should* be read using the csv module because some fields contain semicolons. For example, the "Lokalamnet" (location) for observation 13101005 in `amiral.csv` is given as "Skottsunslätten; Njurunda." If you split rows at semicolons, you will mistakenly divide the row an extra time, leading to incorrect interpretation. The csv module takes into account that certain fields may contain semicolons and uses the protection of quotation marks".

The first row has column headers, followed by observation data, with one observation per row. The columns you need are:

- Artnamn (eng: species name)
- Antal (eng: number/amount)
- Ost och Nord (eng: east/ north)
- Slutdatum (eng: end date)

Antal represents the number of observed individuals in an observation. Some observations have a note ("note-rad") instead of a number. It is suggested to translate this to 1.

Ost and *Nord* provide coordinates in the coordinate system RT 90, indicating positions in meters from a meridian about 15° east of Greenwich and the equator.

Coordinates Ost 1662355 and Nord 6573555 roughly mean "1662 km east of the central meridian and 6573 km north." The central meridian is a kind of midpoint in Sweden, about 15 degrees east of Greenwich. In the RT 90 coordinate system, the northern coordinates are approximately 7585000 for Abisko and 6164000 for Ystad.

Observations have start and end dates since they are sometimes collected over time. For butterfly observations, it is sufficient to use only the *Slutdatum* (End Date). Note that some *Startdatum* (Start Date) values are strangely early. Dates are given in the format 2023-06-01 (ISO standard, "first of June 2023"). It may be practical to use the Python `datetime` module to read and work with dates.

Grading

Details in the general instructions in *INDU–Individual Programming Project*.